

# Lavender Ayako

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## Professional Summary

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Dedicated and analytical Graduate Electrical and Electronics Engineer specializing in power systems, renewable energy integration, solar systems, and machine design. I have a Masters degree in Mathematical Science with an aim of AI-powered mathematical modeling of Complex systems in Engineering field especially Bio-gas systems . I have a foundation in simulation, system modeling, and circuit analysis with experience in HVDC systems and distributed generation. Seeking opportunities to apply engineering expertise in energy, automation, and intelligent systems projects.

## Work Experience

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### Greenium Energy Limited, Bondo, Kenya

2025

#### *Assistant Innovation Officer*

- Installed Biogas for vulnerable women in the village and local primary schools
- Activists in green energy climate solution
- Installed and learnt electrical and electronic appliances related to solar power and power grid systems.
- Provided innovative technical solutions aligned with customer specifications and system ratings. N
- Coordinated with the Chief Electrical Engineer on inventory, spare parts, and maintenance planning.
- Assisted in developing operational plans compliant with safety, legislative, and technical standards.

### KENGEN – Sondu Miriu Power Plant Ltd, Kisumu, Kenya

2024

#### *Electrical Engineering Intern*

- Assisted in maintenance and testing of transformers, circuit breakers, and switchgear.
- Performed fault detection and diagnostic analysis on generating systems.
- Supported substation maintenance planning and equipment reporting.
- Repaired generators and power outlets within the generating station.

### Nambale General Traders, Siaya, Kenya

2023–2024

#### *Industrial Attachee*

- Gained hands-on experience in metal manufacturing including nails, barbed wire, and fencing wire.
- Assisted in operation of wire drawing and nail-making machines.
- Participated in quality control, machine setup, troubleshooting, and routine maintenance.
- Followed strict safety procedures and met daily production targets.

### IEEE Kenya Section, Ugunja, Kenya

2023–Present

#### *IEEE Volunteer/Mentor – Girls in STEM Mentorship Program*

- Mentored high school girls to pursue STEM education and engineering careers.
- Conducted workshops on career guidance, university preparation, and technical skills.
- Collaborated with IEEE members on outreach programs promoting women in engineering.

### Machakos University, Machakos, Kenya

2023

#### *Student Assistant – Power Systems Laboratory*

- Guided students during experiments on synchronous machines and power flow.
- Demonstrated safe laboratory practices and proper equipment handling.

## Education

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### University of Capetown

Sept 2025-Dec 2027

#### Masters of Science in Mathematical Science(Integrating Statistics with Engineering)

- Thesis: Quantifying Impact of extreme wind speed on wind turbine performance using Generalized Extreme Value (GEV): A case of South African Wind farms.

- Key modules: Mathematical problem solving, Sage Math, Python, Verified Computing, Quantum Computing, Ordinary Differential Equations, Statistical Inference, Bayesian Inference, Fluid Dynamics, Material Modelling, Solitons

### **Machakos University, Kenya**

Bachelor of Science in Electrical and Electronics Engineering

**2020–2025**

#### **Grade: First Class Honors**

- Key Modules: Power Systems, HVDC Transmission, Machine Design, Renewable Energy Integration, Control Systems

### **Projects**

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#### **Biogas Installation for Rural community in Siaya, Kenya**

**2025**

Aided in installation of Biogas to vulnerable women in the community to promote green energy solution.

#### **GSM-Based Automatic Fault Detection of Underground Cables**

**2025**

Developed a GSM-enabled monitoring system for real-time detection and location of underground cable faults.

#### **50 kW DC Generator System for Home Use**

**2024**

Designed and simulated a standalone DC generator system with energy storage for domestic applications.

### **Awards and Achievements**

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- Awarded by IEEE for advocating for Women in STEM inclusion
- Awarded the best Mathematical Innovator by African Women Mathematics Association during International Maths Day
- Awarded a full MasterCard Foundation Scholarship to pursue Master's degree at University of Capetown South Africa
- Awarded a 3 year Chancellor Scholarship program due to an outstanding performance in STEM to pursue my Bachelor's degree at Machakos University
- First female Chairperson, Machakos University Election Body (2024) and also the first chairperson to conduct a free and fair election since the university elections started
- Dean's List – Best Overall Electrical Engineering Student (2023)
- Solar Energy Measurement Competition – University Team Lead (2022)
- IEEE awarded Best volunteer in Girls in Stem Mentorship Activists

### **Training & Certifications**

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- Python, Pytorch, SageMath, Leadership certificates, Proficiency
- STCW Certification
- Certificate of Competency (CoC)
- Occupational Health and Safety Induction (KENGEN)
- MATLAB for Electrical Engineers – Coursera
- Renewable Energy Systems Fundamentals – Greenium

### **Skills**

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**Technical Tools:** MATLAB, Simulink, AutoCAD Electrical, PSCAD, Proteus, Latex

**Programming:** Python, C, Arduino

**Core Areas:** Power Systems, Renewable Energy Integration, HVDC Design, Control Systems, Machine Analysis

### **Referees**

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Available upon request